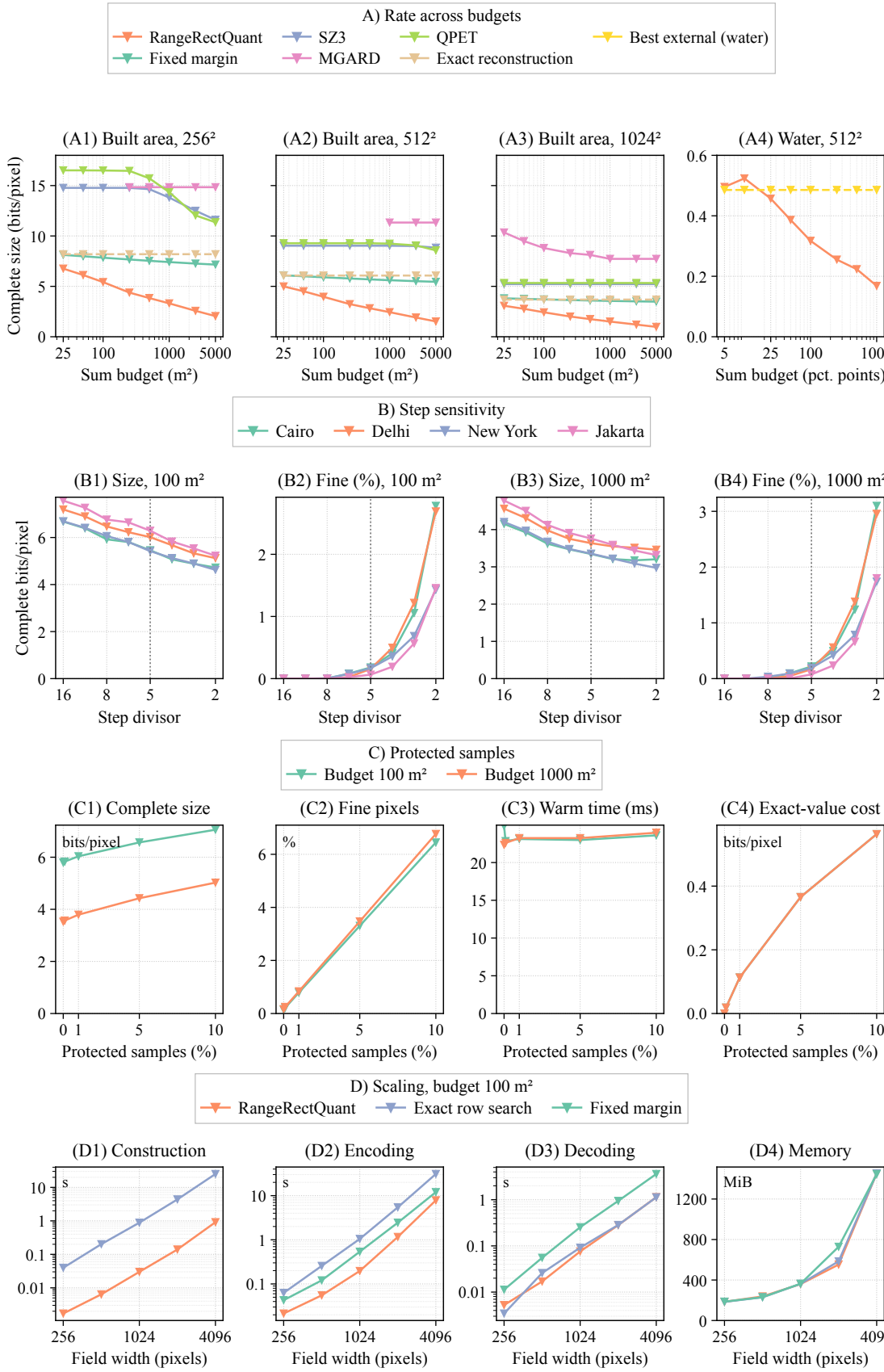
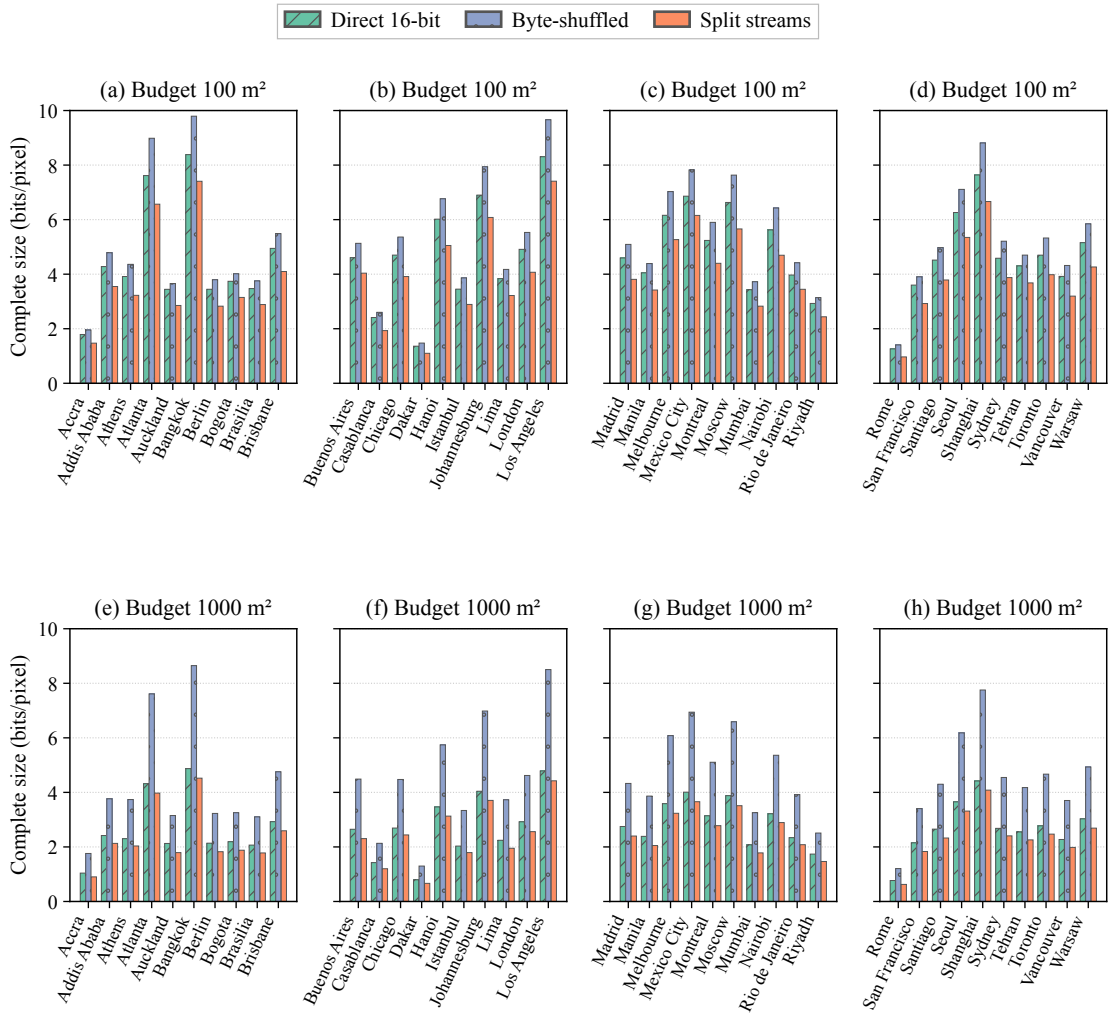
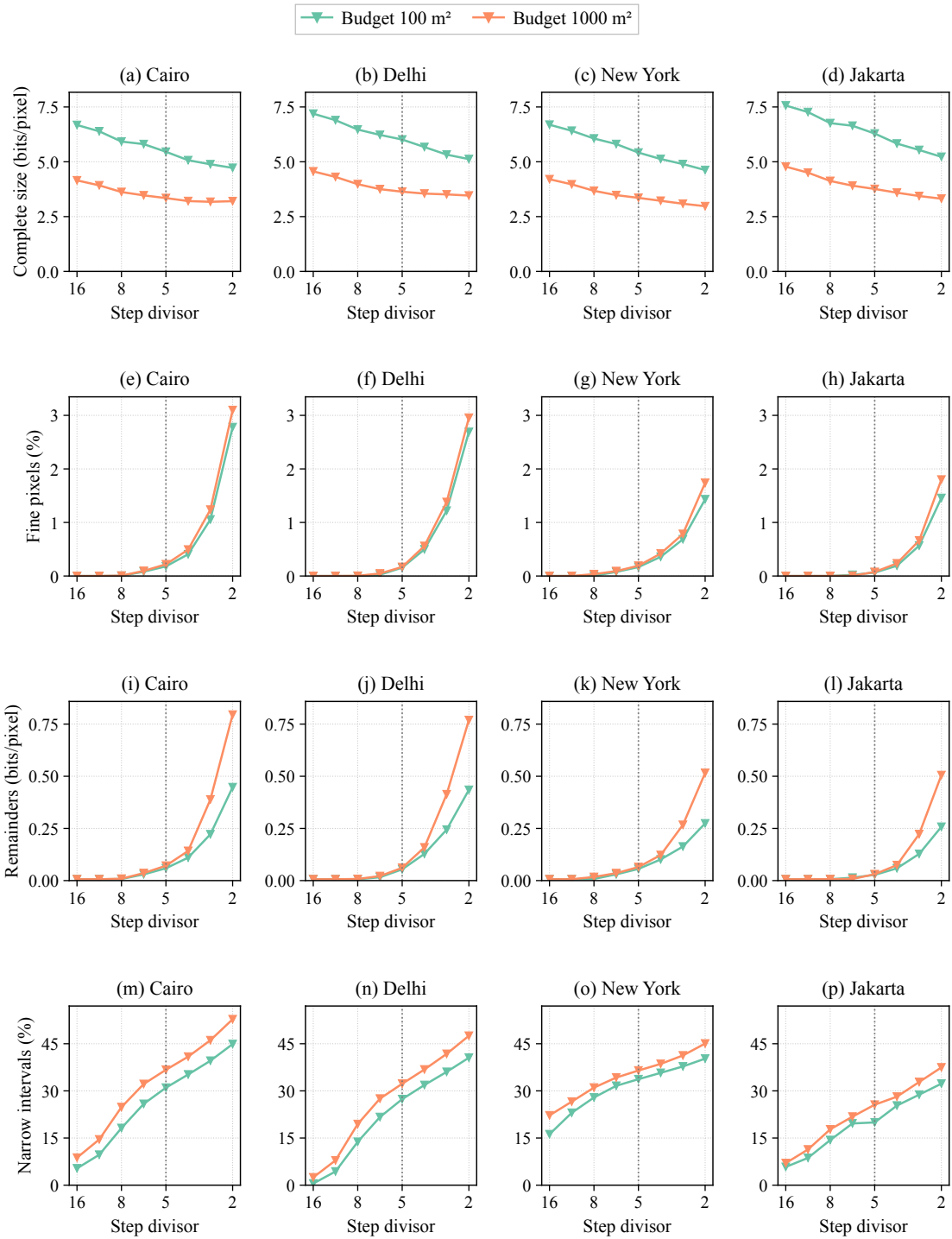
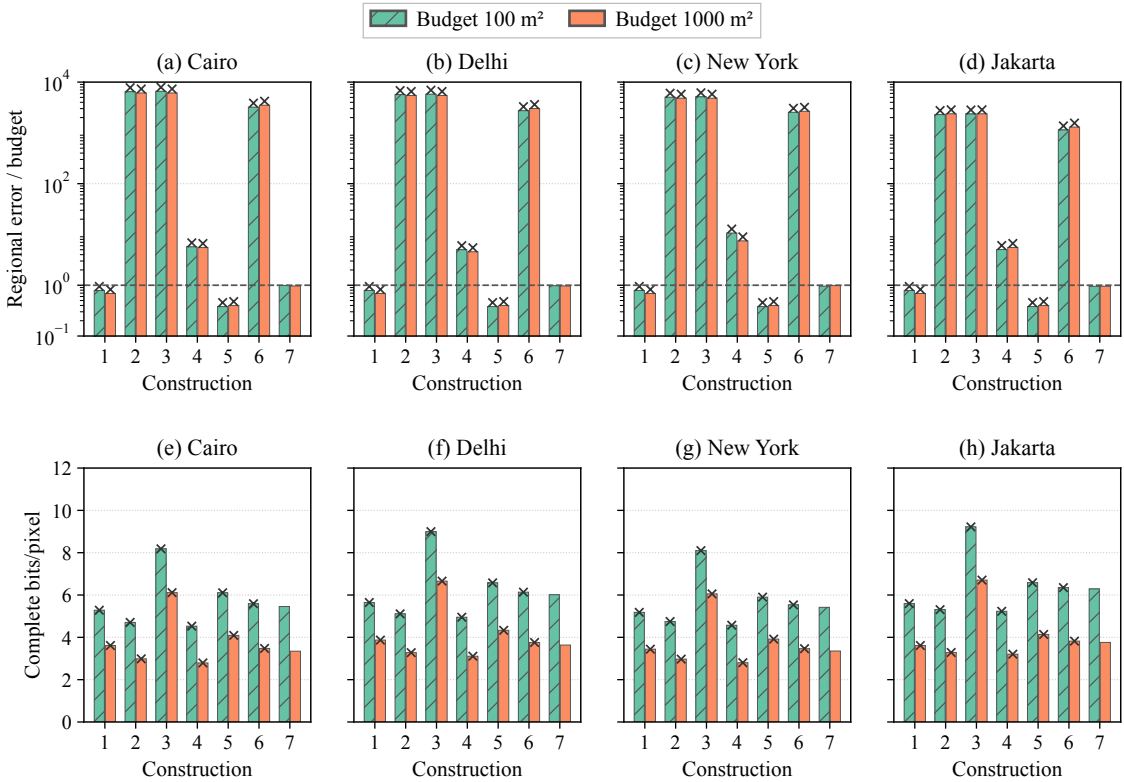

A) Rate across budgets
RangeRectQuant
SZ3
QPET
Best external (water)
Fixed margin
MGARD
Exact reconstruction
(A1) Built area, 256²
(A2) Built area, 512²
(A3) Built area, 1024²
(A4) Water, 512²
Complete size (bits/pixel)
Sum budget (m²)
Sum budget (pct. points)
B) Step sensitivity
Cairo
Delhi
New York
Jakarta
(B1) Size, 100 m²
(B2) Fine (%), 100 m²
(B3) Size, 1000 m²
(B4) Fine (%), 1000 m²
Complete bits/pixel
Step divisor
C) Protected samples
Budget 100 m²
Budget 1000 m²
(C1) Complete size
(C2) Fine pixels
(C3) Warm time (ms)
(C4) Exact-value cost
bits/pixel
%
Protected samples (%)
D) Scaling, budget 100 m²
RangeRectQuant
Exact row search
Fixed margin
(D1) Construction
(D2) Encoding
(D3) Decoding
(D4) Memory
s
MiB
Field width (pixels)



Forty 512² locations, alphabetical order; each triplet encodes exactly the same field.

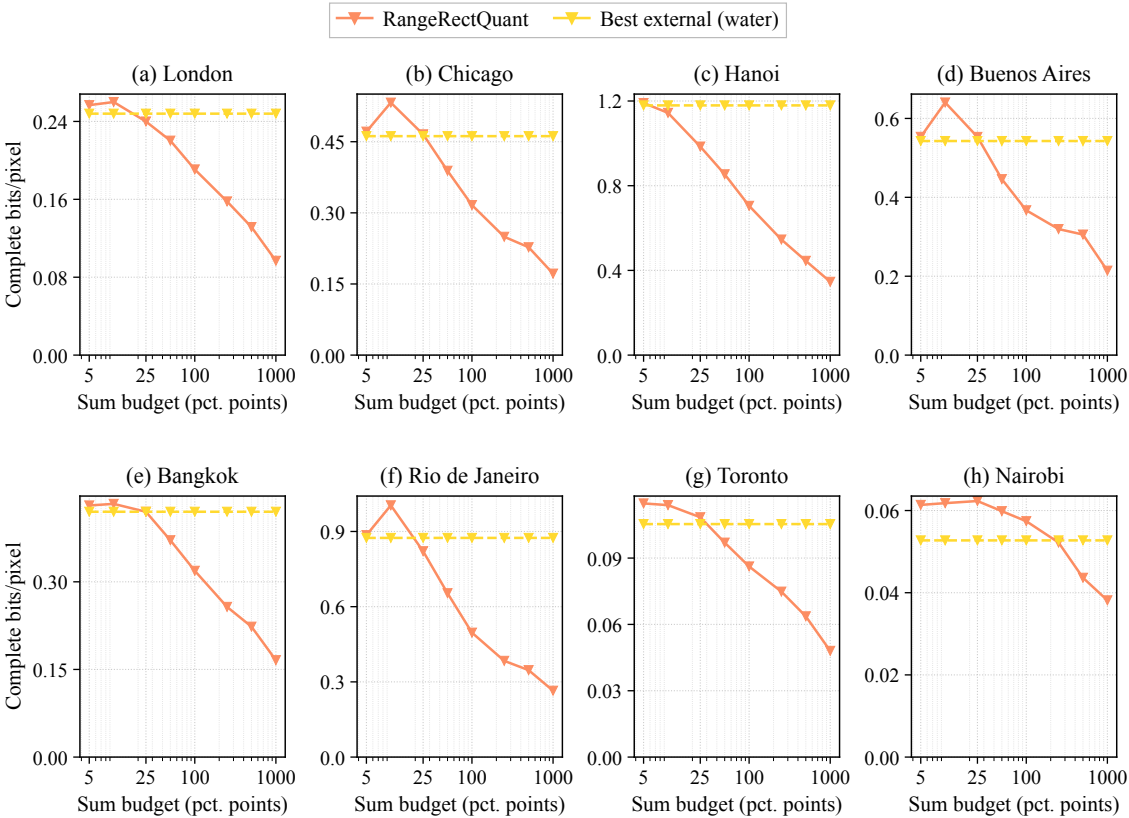


Four earlier holdouts. Dotted lines: default divisor five. Narrow intervals bound the possible fine pixels.

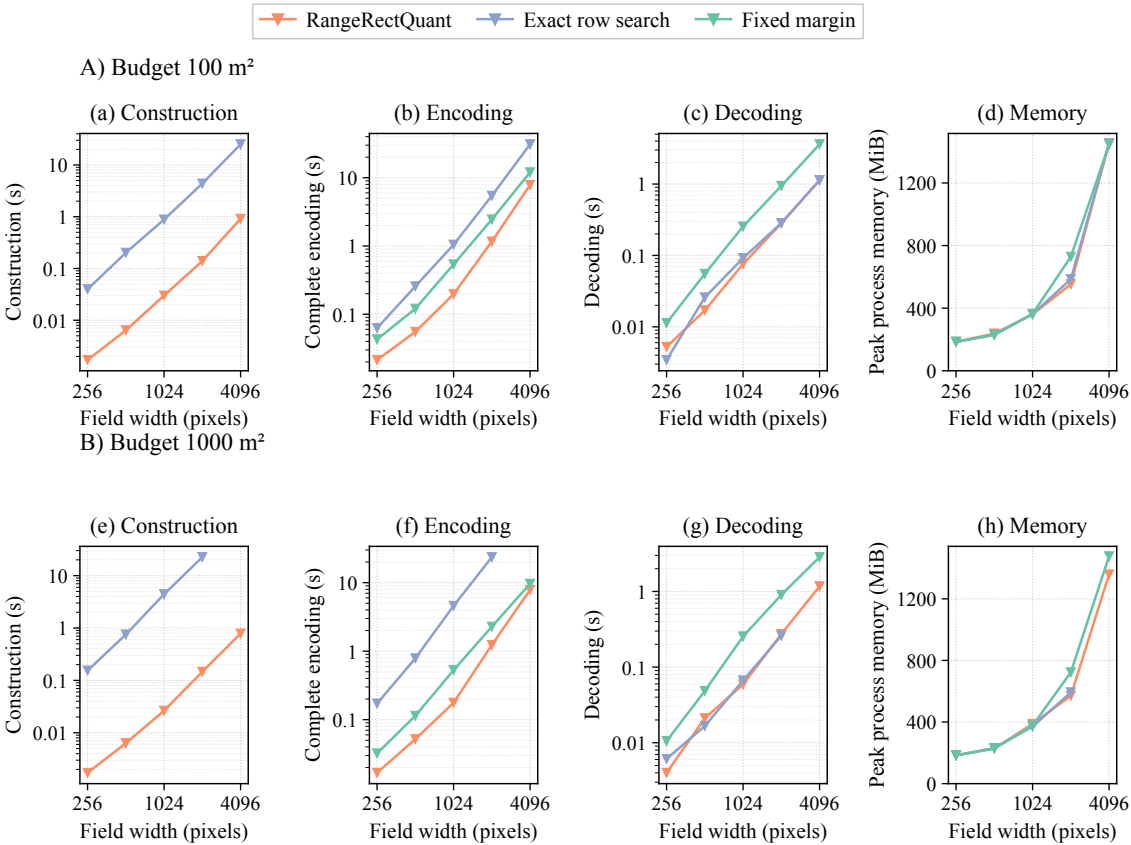


1: Stable sigma–delta 2: + clipping 3: + clipping + integer rounding
4: Projected sigma–delta 5: Prefix rounding 6: + clipping 7: RangeRectQuant

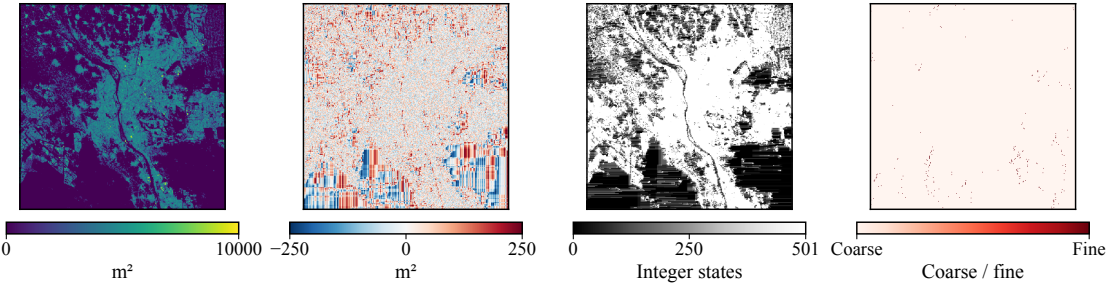
×: integer, range or regional violation. Dashed line: regional budget. All 56 decoded files shown.



Eight native 512² water-occurrence fields; all eight budgets. Each panel has an independent y-axis.
Complete-file rates include masks and metadata. Budgets are percentage-point sums.



Three warm repetitions; whole-process memory. Exact search: 4096², 1000 m² worker timeout at 180 s.
Construction measurements cover RangeRectQuant and exact row search.


(a) Built area
(b) Prefix error
(c) Allowed states
(d) Fine pixels
0
10000
m²
−250
0
250
m²
0
250
501
Integer states
Coarse
Fine
Coarse / fine